Dr. Torsten Busch
Tel.: +49 (0) 69/257203 012
Fax: +49 (0) 69/257203 999
Email: torsten.busch@curiaglobal.com

Curia Germany GmbH
Process Research & Development
Industriepark Hoechst, D569
D-65926 Frankfurt am Main

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Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
  - three syntheses done with 0.02M sulfuric acid
    - first product not representative, not used further
    - second and third batch converted (see below)
  - four syntheses done with 0.10M sulfuric acid
    - very similar observations and outputs
    - products of these have been combined/homogenized
    - part of the combined lots has been converted

- Canadian Rockies Hemp arrived and cleared customs
  - several standard synthesis batches finished, seven of these (11.2 g) combined and homogenized
  - additional experiments done with higher concentrated sulfuric acid (1.5%, 2.0%, 5.0%)
  - one experiment performed under stirring, using cut up hemp fibers, product is finer and more homogeneous
  - one experiment performed with sulfuric acid pre-treatment of hemp (95 °C)

- Larger vessel reactivated, production ongoing (approx. 15 g output per batch)

- Typical moisture content 5% by KFT. Further drying at 100 °C under nitrogen stream decreases KFT result to 2%. Relevance unclear – being tested in step 2.

- Corrosion/cleaning test for large-scale equipment in progress, first formal meeting with pilot plant staff held

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
  - biochar ground manually for first experiment (TB1131)
  - laboratory mill used for second experiment (TB1132)
  - products not yet characterized

- no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)
    - better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h: minimal yield (MB2947A)
    - 800 °C, 2 h: minimal yield (MB2947B)
    - 800 °C, 3 h: no output (MB2947C)
    - ➔ Inertion apparently insufficient and material likely incinerated
    - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
    - 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
    - 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
    - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
        - ➔ smoldering observed upon contact of the cold material with air (see video)
    - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
        - ➔ material wetted with water before/during removal from the apparatus

- Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)

- 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
    - 800 °C, 1 h, standard inertion (repeat of MB2955A): 90 mg output (MB2955A2)
    - 800 °C, 2 h, standard inertion: 80 mg output (TB1136)

- 2 g of combined biochar milled with 4 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: 80 mg output (MB2962A)
    - 800 °C, 2 h, standard inertion: 90 mg output (MB2962B)

- 0.6 g of biochar (cut up, hydrolyzed under stirring) milled with 0.6 g of KOH and used for two batches on 0.2 – 0.3 g scale
    - 800 °C, 1 h, standard inertion: 70 mg output (MB2963A)
    - 800 °C, 2 h, standard inertion: 120 mg output (MB2963B)

- 2 g of combined biochar milled with 8 g of KOH and used for several batches on 0.25 g scale
    - 800 °C, 1 h, standard inertion: 60 mg output (TB1137-1)
    - 800 °C, 2 h, standard inertion: 60 mg output (TB1137-2)
    - 800 °C, 1 h, standard inertion: 70 mg output (TB1137-3)

- 2 g of biochar lot MB2959 milled with 12 g of KOH and used for several batches
    - 800 °C, 1 h, standard inertion: 24 mg output (MB2965A)
    - 800 °C, 2 h, standard inertion: 13 mg output (MB2965B)

- 800 °C, 2 h, standard inertion (oven B): 7 mg output (MB2965B2)
- 2 g of biochar lot MB2959 milled with 8 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion (oven B): 52 mg output (MB2966A)
  - 800 °C, 2 h, standard inertion: 46 mg output (MB2966B)

- 1.2 g of biochar lot MB2964 milled with 4.8 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion (oven B): 57 mg output (MB2967A)
  - 800 °C, 2 h, standard inertion: 42 mg output (MB2967B)

- 1.4 g of biochar lot MB2957 (from 1.5% H_2SO_4) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion (oven B): 63 mg output (MB2970A)
  - 800 °C, 2 h, standard inertion: 19 mg output (MB2970B)

- 1.4 g of biochar lot MB2958 (from 2% H_2SO_4) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion (oven B): 54 mg output (MB2971A)
  - 800 °C, 2 h, standard inertion: 66 mg output (MB2971B)

- 1.4 g of biochar lot MB2961 (from 5% H_2SO_4) milled with 2.1 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion (oven B): 88 mg output (MB2972A)
  - 800 °C, 2 h, standard inertion: 57 mg output (MB2972B)

- 1.4 g of biochar lot MB2959 milled with 2.1 g of KOH and used for two identical batches (reproducibility experiment)
  - 800 °C, 1 h, standard inertion (oven B): 42 mg output (MB2973A)
  - 800 °C, 1 h, standard inertion: 90 mg output (MB2973B)

- 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
  - milling time 5 min, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2974A)
  - milling time 1 min, 800 °C, 1 h, standard inertion: 81 mg output (MB2974B)

- 1.5 g of biochar lot MB2959 milled with 2.3 g of KOH and used for two batches (milling time experiment)
  - milling time 5 min, 800 °C, 2 h, standard inertion (oven B): 59 mg output (MB2975A)
  - milling time 1 min, 800 °C, 2 h, standard inertion: 80 mg output (MB2975B)

- 2.0 g of biochar lot MB2959 milled with 3.0 g of KOH and used for two batches (pellet pressing experiment)
  - 0.6 g pellet, 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2976A)
  - 0.6 g pellet, 800 °C, 2 h, standard inertion: 108 mg output (MB2976B)

- Biochar/KOH mixture from MB2976 used for new reproducibility experiment (analogous to MB2973)
    - 800 °C, 1 h, standard inertion (oven B): 83 mg output (MB2979A)
    - 800 °C, 1 h, standard inertion: 104 mg output (MB2979B)

- 2.5 g of biochar lots MB2959/2956 milled with 3.8 g of KOH for 10/20 min and used for four batches (pellet pressing experiment)
    - 0.6 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 78 mg output (MB2980A)
    - 0.6 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 95 mg output (MB2980B)
    - 1.3 g pellet, 20 min milling time, 800 °C, 1 h, standard inertion (oven B): 163 mg output (MB2980C)
    - 1.3 g pellet, 10 min milling time, 800 °C, 1 h, standard inertion (oven A): 232 mg output (MB2980D)

- 2.5 g of biochar lot MB2978 milled with 3.8 g of KOH for 10/20 min and used for ten batches
    - 800 °C, 1 h, 10 min milling time, standard inertion (oven A): 95 mg (KJo-0165A)
    - 800 °C, 1 h, 20 min milling time, inertion potentially nonstandard (oven B): 146 mg (KJo-0165B)
    - 800 °C, 1 h, 10 min milling time, 10 °C/min, standard inertion (oven A): 125 mg (KJo-0166A)
    - 800 °C, 1 h, 20 min milling time, 10 °C/min, standard inertion (oven B): 114 mg (KJo-0166B)
    - 800 °C, 4 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 102 mg (KJo-0167A)
    - 800 °C, 4 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 61 mg (KJo-0167B)
    - 900 °C, 2 h, 10 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 23 mg (KJo-0168A)
    - 900 °C, 1 h, 20 min milling time, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 89 mg (KJo-0168B)
    - 800 °C, 1 h, 10 min milling time, 3 °C/min, 2.0 g pellet pressed, standard inertion (oven B): 232 mg (KJo-0169)
    - 800 °C, 1 h, 20 min milling time, 3 °C/min, 2.7 g pellet pressed, standard inertion (oven B): 619 mg (KJo-0170)

- 5 g of biochar lot MB2978 milled with 7.5 g of KOH for 5 min and used for two batches
    - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven B): 642 mg (KJo-0171A)
    - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 128 mg (KJo-0171B)

- 2.5 g of biochar lot MB2978 milled with 2.5 g of KOH for 2.5 min and used for two batches
  - 800 °C, 1 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 243 mg (KJo-0172A)
  - 800 °C, 2 h, 3 °C/min, 0.6 g pellet pressed, standard inertion (oven A): 169 mg (KJo-0172B)

- 3.0 g of biochar lot MB2978 milled with 4.5 g of KOH for 5 min and used for three batches
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A), quartz tube shattered: 710 mg output (KJo-0175)
  - 800 °C, 1 h, 1.5 °C/min, 0.6 g pellet pressed, standard inertion (oven B): 104 mg output (KJo-0176)
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, standard inertion (oven A): 908 mg output (KJo-0177)

- Various combined lots of milled biochar/KOH (1:1.5, lots MB2955/2974/2975/2976) used for two production experiments
  - 800 °C, 1 h, 3 °C/min, 2 x 2.0 g pellets pressed, questionable inertion (oven A): 549 mg output (KJo-0178)
  - 800 °C, 1 h, 3 °C/min, 6 x 0.6 g pellets pressed, standard inertion (oven B): 441 mg output (MB2981)

- Various combined lots of biochar (MB2946/2948/2949/2950/2951/2953/2956/MRa255) milled with KOH (1:1.5) – mixture heated itself moderately during milling
  - 800 °C, 1 h, 3 °C/min, 1.8 + 2.1 g pellets pressed, standard inertion (oven A): 587 mg output (TB1138)
  - 800 °C, 1 h, 3 °C/min, 1.8 + 1.7 g pellets pressed, standard inertion (oven B): 547 mg output (MRa320)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven B): 476 mg output (MB2984)
  - 800 °C, 1 h, 3 °C/min, 1.3 + 1.4 g pellets pressed, standard inertion (oven A): 395 mg output (TB1141)

- 8.5 g biochar (KJo-0173) milled with KOH (1:1.5) – brown powder, quickly turned dark brown, somewhat chunky and hot (steaming) upon contact with air, product thus not perfectly homogeneous
  - 800 °C, 1 h, 3 °C/min, 2 x 1.8 g pellets pressed, standard inertion (oven A): 538 mg output (TB1139)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 464 mg output (TB1140A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 480 mg output (TB1140B)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 421 mg output (MB2988)

- 5.0 g biochar (KJo-0173) milled, sonicated (1 h), milled with KOH (1:1.5) – brown powder, no noteworthy observations
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 464 mg output (MRa323)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 505 mg output (MRa326)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 459 mg output (MRa327)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.5 g pellets pressed, standard inertion (oven B): 528 mg output (MRa328)

- 1.1 g biochar (KJo-0173, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, no noteworthy observations
  - 800 °C, 1 h, 3 °C/min, 2 x 1.5 g pellets pressed, standard inertion (oven B): 434 mg output (MRa329) ➔ shinier, more voluminous product, color shifted to grey/silver

- 7.0 g biochar (MB2982) milled with KOH (1:1.5) – brown powder, turned dark brown, somewhat chunky and hot during/after bottling, manually homogenized
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 505 mg output (MB2990A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 420 mg output (MB2990B)
  - biochar/KOH powder dried in N_2 stream (100 °C, 3 d), 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 524 mg output (TB1146)

- 7.0 g biochar (MB2985, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, turned dark brown, somewhat chunky and hot during/after bottling, manually homogenized
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 637 mg output (MRa331A) ➔ shinier, more voluminous product, color shifted to grey/silver
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 462 mg output (MRa331B) ➔ shinier, more voluminous product, color shifted to grey/silver
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 548 mg output (MB2997A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 452 mg output (MB2997B)

- 7.0 g biochar (MB2985, extra dried at 100 °C) milled with KOH/NaOH (60/40, 1:1.5) – brown powder, stricter inertion after milling thus (?) no self-heating and darkening
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 212 mg output (MRa333A) ➔ shinier, more voluminous product, color shifted to grey/silver
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 258 mg output (MRa333B) ➔ shinier, more voluminous product, color shifted to grey/silver

- 1:1 mixture of powders from MRa331 (biochar:KOH 1:1.5) and MRa333 (biochar:KOH:NaOH 1:0.9:0.6) ➔ KOH:NaOH 0.8:0.2, used for two experiments
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven A): 476 mg output (MRa334A) ➔ shinier, more voluminous product, color shifted to grey/silver
  - 800 °C, 1 h, 3 °C/min, 2 x 1.1 g pellets pressed, standard inertion (oven B): 394 mg output (MRa334B) ➔ shinier, more voluminous product, color shifted to grey/silver

- 7.0 g biochar (MB2982) milled with KOH/NaOH (60/40, 1:1.5) – brown powder, stricter inertion after milling thus (?) no self-heating and darkening
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 225 mg output (MB2993A) ➔ shinier, more voluminous product, color shifted to grey/silver
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 189 mg output (MB2993B) ➔ shinier, more voluminous product, color shifted to grey/silver

- 7.0 g biochar (MB2983, extra dried at 100 °C) milled with KOH (1:1.5) – brown powder, part of the mixture started smoldering during bottling and was lost, the bottled material was not affected and remained fine brown powder
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 536 mg output (MB2995A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 457 mg output (MB2995B)
  - biochar/KOH powder dried in N_2 stream (100 °C, 3 d), 800 °C, 1 h, 3 °C/min, 2 x 1.3 g pellets pressed, standard inertion (oven B): 355 mg output (TB1147)

- 2.8 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (15 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
  - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven A): 613 mg output (TB1144A)
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven B): 355 mg output (TB1144B)

- 3.4 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (25 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 676 mg output (MB2999A)
  - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 400 mg output (MB2999B)

- 3.4 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (35 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 493 mg output (MB3001A)
  - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 478 mg output (MB3001B)

- 5.5 g 2% wet biochar (MB2986) milled with KOH (1:1.5) in ball mill (35 Hz, 80 sec) – brown powder, appeared homogeneous and was bottled without apparent reaction with air
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 574 mg output (MB3003A) ➔ powder used fresh

- 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 435 mg output (MB3003B) ➔ powder used fresh
- 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): 592 mg output (MB3004A) ➔ powder used after standing for a day
- 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): 406 mg output (MB3004B) ➔ powder used after standing for a day

- 4.6 g biochar (KJo-0174) milled with KOH (1:1.5) in ball mill (35 Hz, 10 min) – brown powder, appeared homogeneous, limited decoloring noticed during bottling under nitrogen
  - 800 °C, 1 h, 3 °C/min, 1.7/1.8 g pellets pressed, standard inertion (oven A): 531 mg output (TB1151A)
  - 800 °C, 1 h, 3 °C/min, 2.7 g powder, standard inertion (oven B): 450 mg output (TB1151B)

- 4.3 g biochar (KJo-0174/MB2982/MB2989) milled with KOH (1:1.5) in ball mill (35 Hz, 30 min) – brown powder, appeared homogeneous, limited decoloring noticed during bottling under nitrogen
  - 800 °C, 1 h, 3 °C/min, 2 x 1.6 g pellets pressed, standard inertion (oven A): output tbd (MB3007A) ➔ second pellet disintegrated, only parts used for reaction (attempted twice)
  - 800 °C, 1 h, 3 °C/min, 2.6 g powder, standard inertion (oven B): output tbd (MB3007B)

- The following lots were combined and ground briefly, to generate one large batch (TB1142, 7.7 g)

    - MB2976 A/B: approx. 100 mg
    - MB2980 A: approx. 50 mg
    - MB2980 C/D: approx. 300 mg
    - KJo-0166 B: approx. 100 mg
    - KJo-0167 B: approx. 50 mg
    - KJo-0168 B: approx. 70 mg
    - KJo-0171 A: approx. 600 mg
    - KJo-0171 B: approx. 100 mg
    - KJo-0172 A: approx. 220 mg
    - KJo-0175: approx. 600 mg
    - KJo-0176: approx. 80 mg
    - KJo-0177: approx. 800 mg
    - KJo-0178: approx. 500 mg
    - MB2981: approx. 400 mg
    - TB1138: approx. 500 mg
    - MRa320: approx. 500 mg
    - MB2984: approx. 400 mg
    - TB1139: approx. 500 mg
    - TB1140A: approx. 400 mg
    - TB1140B: approx. 400 mg
    - MRa323: approx. 400 mg
    - TB1141: approx. 300 mg

4. Graphene analysis

- SEM (and partially Raman) results available for
  - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products), MB2965B, MB2965B2, MB2966A/B, MB2967A/B, MB2970A/B, MB2971A/B, MB2972A/B, MB2973A/B, MB2974A/B, MB2975A/B, MB2976A/B, MB2979A/B, MB2980A/B/C/D, KJo-0165A/B, KJo-0166A/B, KJo-0167A/B, KJo-0168A/B, KJo-0169, KJo-0170, KJo-0171A/B, KJo-0172A/B, KJo-0175, KJo-0176, KJo-0177, KJo-0178, KJo-0171A GR, MB2981A, TB1138, MB2984, MRa320, TB1139, TB1140A, TB1140B, MB2988, MRa323, TB1141, MRa326, MRa327, MRa328, MB2990A, MB2990B, MRa329, MRa331A, MRa331B, MRa333A, MRa333B, MRa334A, MRa334B, MRa2993A, MRa2993B, MB2995A, MB2995B, MB2997A, MB2997B, TB1144A, MB1144B, MB2999A, MB2999B, TB1146, TB1147, MB3001A, MB3001B, MB3003A, MB3003B, MB3004A, MB3004B (step 2 products)
  - 900561-500MG (Sigma-Aldrich reference)
  - 924458-500MG (Sigma-Aldrich reference "Single-layer graphene sheets for battery, Bio-sourced")

- TEM results available for
  - MB2955A2 (step 2 product)
  - MB2976A (step 2 product)
  - TB1137-1 (step 2 product)

5. Currently open questions

- Patent: new process/patent
    - So far mainly focusing on pellet (weak)
    - Needs to brainstorm/think about other improvements/new ideas

- Characteristics of the Graphene
    - BET results
    - RAMAN results
    - Electrical conductivity
    - Volume fraction of mesoporosity

- Starting discussion on next size commercial production
    - Group call pending

- Is partial replacement of KOH by NaOH or LiOH beneficial?

- Does the shiny, voluminous material (as obtained from MRa329) have desirable properties?
    - Which parameters lead to the formation of this material?

- Does the use of a ball mill lead to different results compared to the previously used "coffee grinder" type mill?

- What effect do different milling parameters have on the properties of the product?

- How to mill and press larger quantities of biochar during scaleup?

6.Questions Answered

- Are we able to reproduce Dr Li's extraction
    - By SEM: yes (based on SEM at Dr Li's SEM)
    - By TEM: result needed to confirm

- Are we able to produce 3-5g of Graphene
    - Done (TB1142, 7.7 g)

- What are the next steps to Scale-up
    - Step 1: Decision on reaction vessel, corrosion/cleaning test
    - Step 2: Reproduction of previous results with rotating oven

- Rotating oven
    - Ordered (delivery early November)

Experiment overview – Step 1

Experiment	Reactor		Raw material	Acid				T	t	pressure			Wash		Output	
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.4 bar	742 g	Water	2.6 g	
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.5 bar	647 g	Water	1.2 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	9.5 bar	500 g	Water	1.3 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	->	10.5 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	11.0 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	->	10.5 bar	500 g	Water	0.8 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	10.9 bar	569 g	Water	1.6 g	Combined and homogenzied as lot no MB2946X
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	->	13.2 bar	577 g	Water	1.4 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.6 bar	555 g	Water	1.5 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	->	13.5 bar	578 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	->	12.7 bar	611 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	->	9.7 bar	593 g	Water	1.8 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.8 bar	6 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	->	12.6 bar	475 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output	Drying	KFT
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar	->	10.2 bar	538 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h		->	11.1 bar	534 g	Water	1.6 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar	->	11.2 bar	575 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar	->	12.8 bar	488 g	Water	1.7 g		
Comment	some foreign material observed and removed during filtration																
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar	->	10.4 bar	558 g	Water	1.8 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	12.5 bar	548 g	Water	1.9 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar	->	8.5 bar	589 g	Water	1.5 g		
Comment																	
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar	->	8.7 bar	562 g	Water	1.8 g		
Comment																	
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h				3957 g	Water	14.9 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar	->	10.5 bar	300 g	Water	0.6 g		
Comment	Reaction performed with stirring																
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar	->	16.1 bar	563 g	Water	1.7 g		
Comment	low amount of foreign material observed and removed during filtration																
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	8.8 bar	310 g + 603 g	Water	1.3 g		
Comment	Hemp boiled with 1% H_2SO_4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid																
MB2978	AV5	76.3 g	Canadian Rockies Hemp	822.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	22 h				4112 g	Water	15.4 g		
Comment																	
KJo-0173	AV5	77.1 g	Canadian Rockies Hemp	942.1 g	1.00%	0.10 M	Sulfuric acid	185 °C	24 h				4283 g	Water	16.5 g	40 °C	4.9%
Comment																100 °C	2.3%
KJo-0174	AV5	77.1 g	Canadian Rockies Hemp	945.0 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4384 g	Water	15.8 g	40 °C	3.7%
Comment																	
MB2982	AV5	77.1 g	Canadian Rockies Hemp	999.5 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4258 g	Water	14.9 g	40 °C	5.3%
Comment																	
MB2983	AV5	80.5 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	190 °C	24 h				4232 g	Water	16.6 g	40 °C	5.5%
Comment																100 °C	1.7%

Experiment	Reactor	Raw material			Acid			T	t	pressure	Wash		Output	Drying	KFT
MB2985	AV5	79.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	200 °C	24 h		4241 g	Water	15.4 g	40 °C	3.7%
Comment														100 °C	1.9%
MB2986	AV5	79.9 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4128 g	Water	16.0 g	60 °C	2.7%
Comment														100 °C	1.7%
MB2989	AV5	80.2 g	Canadian Rockies Hemp	999.3 g	1.00%	0.10 M	Sulfuric acid	180 °C	23 h		4314 g	Water	18.2 g	40 °C	5.9%
Comment															
MB2991	AV5	81.3 g	Canadian Rockies Hemp	999.3 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4223 g	Water	19.1 g	40 °C	5.7%
Comment															
MB2992	AV5	81.5 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	160 °C	23.5 h		4262 g	Water	19.5 g	40 °C	6.2%
Comment															
MB2994	AV5	80.8 g	Canadian Rockies Hemp	999.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4159 g	Water	20.2 g	40 °C	5.2%
Comment															
MB2996	AV5	82.7 g	Canadian Rockies Hemp	999.7 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4210 g	Water	20.0 g	40 °C	4.6%
Comment															
MB2998	AV5	84.0 g	Canadian Rockies Hemp	999.6 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4263 g	Water	19.0 g	40 °C	
Comment	Incuded nickel and steel material samples for corrosion/cleaning tests														
MB3000	AV5	82.5 g	Canadian Rockies Hemp	1000.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4431 g	Water	18.9 g	40 °C	
Comment															
MB3002	AV5	80.0 g	Canadian Rockies Hemp	999.9 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4245 g	Water		40 °C	
Comment															
MB3005	AV5	81.4 g	Canadian Rockies Hemp	999.4 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h		4280 g	Water		40 °C	
Comment															
MB3006	AV5	80.8 g	Canadian Rockies Hemp	1000.1 g	1.00%	0.10 M	Sulfuric acid								
Comment															

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
TB1131	A	1.2 g	MB2933	1.2 g	KOH	(90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										(+ water)		(atm. Pressure)		
TB1132	A	1.3 g	MB2935	1.3 g	KOH	(90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder										(+ water)		(atm. Pressure)		
TB1133	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1134	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2946	A	1.0 g	MB2946X	1.0 g	KOH	(90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2947A	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded				
Comment		ground biochar: brown powder; see temperature trend														
MB2947D	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g	10% HCl	100 °C	N2 stream	38 mg
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-1	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2952C2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2955A	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	7 mg
Comment		Quartz tube shattered during experiment										(+ water)		(atm. Pressure)		
MB2955A2	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1136	A	0.26 g	MB2946X	0.39 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	14 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962A	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962B	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963A	A	0.22 g	MB2960	0.22 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963B	A	0.29 g	MB2960	0.29 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-1	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-2	A	0.28 g	MB2946X	1.13 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	2 h	23 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-3	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N_2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2965A	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N_2 stream	0.02 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B2	B	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.01 g
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2966A	B	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N_2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2966B	A	0.26 g	MB2959	1.04 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967A	B	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2967B	A	0.26 g	MB2964	1.04 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	20 g	10% HCl	100 °C	N_2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970A	B	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2970B	A	0.24 g	MB2957	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.02 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971A	B	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2971B	A	0.24 g	MB2958	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972A	B	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2972B	A	0.24 g	MB2961	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N_2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.04 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2973B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2974B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N_2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N_2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2975A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2975B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2976A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2976B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.11 g
Comment		ground biochar (brown powder) compacted to pellet; see temperature trend										(+ water)		(atm. Pressure)		
MB2979A	B	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2979B	A	0.24 g	MB2959	0.36 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2980A	B	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980B	A	0.24 g	MB2959/2956	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980C	B	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.16 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
MB2980D	A	0.52 g	MB2959/2956	0.78 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	21 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0165A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0165B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.15 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		
KJo-0166B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	10 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.11 g
		ground biochar: brown powder										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
KJo-0167A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	4 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0167B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	4 h	11 g	10% HCl	100 °C	N2 stream	0.06 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168A	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	900 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.02 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0168B	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	900 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0169	B	0.80 g	MB2978	1.20 g	KOH	(90%)	mill (10 min)	N2	3 °C/min	800 °C	1 h	30 g	10% HCl	100 °C	N2 stream	0.23 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0170	B	1.07 g	MB2978	1.61 g	KOH	(90%)	mill (20 min)	N2	3 °C/min	800 °C	1 h	38 g	10% HCl	100 °C	N2 stream	0.62 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0171A	B	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.64 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0171B	A	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.13 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172A	B	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	11 g	10% HCl	100 °C	N2 stream	0.24 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0172B	A	0.50 g	MB2978	0.50 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.17 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0175	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.71 g
		ground biochar (brown powder) compacted to two pellets of equal size, quartz tube shattered										(+ water)		(atm. Pressure)		
KJo-0176	B	0.24 g	MB2978	0.36 g	KOH	(90%)	mill (5 min)	N2	1.5 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.10 g
		ground biochar (brown powder) compacted to pellet										(+ water)		(atm. Pressure)		
KJo-0177	A	1.60 g	MB2978	2.40 g	KOH	(90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	61 g	10% HCl	100 °C	N2 stream	0.91 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
KJo-0178	A	1.60 g	MB2955/74/75/76	2.40 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.55 g
		ground biochar (brown powder) compacted to two pellets of equal size										(+ water)		(atm. Pressure)		
MB2981	B	1.44 g	MB2955/74/75/76	2.16 g	KOH	(90%)	mill (1-5 min)	N2	3 °C/min	800 °C	1 h	60 g	10% HCl	100 °C	N2 stream	0.44 g
		ground biochar (brown powder) compacted to six pellets of equal size										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Appearance	Structure
TB1138	A	1.54 g	various	2.31 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	59 g 10% HCl	100 °C N2 stream	0.59 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	38%	brittle	
MRa320	B	1.43 g	various	2.14 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	53 g 10% HCl	100 °C N2 stream	0.55 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	38%	brittle	
MB2984	B	1.40 g	various	2.10 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	53 g 10% HCl	100 °C N2 stream	0.48 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	34%	brittle	
TB1139	A	1.39 g	KJo-0173	2.08 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	50 g 10% HCl	100 °C N2 stream	0.54 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	39%	brittle	
TB1140A	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	49 g 10% HCl	100 °C N2 stream	0.46 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	36%	brittle	
TB1140B	B	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	50 g 10% HCl	100 °C N2 stream	0.48 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	38%	brittle	
MRa323	A	1.06 g	KJo-0173	1.60 g KOH (90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.44 g	black/grey	porous chunks
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	41%	brittle	
TB1141	A	1.08 g	various	1.62 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	30 g 10% HCl	100 °C N2 stream	0.40 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	37%	brittle	
MRa326	A	1.28 g	KJo-0173	1.92 g KOH (90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.51 g	black/grey	porous chunks
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	39%	brittle	
MRa327	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g	black/grey	porous chunks
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	44%	brittle	
MB2988	B	1.04 g	KJo-0173	1.56 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.42 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	40%	brittle	
MRa328	B	1.20 g	KJo-0173	1.80 g KOH (90%)	mill (3 min)	N2	3 °C/min	800 °C	1 h	45 g 10% HCl	100 °C N2 stream	0.53 g	black/grey	porous chunks
		ground biochar (sonicated, brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	44%	brittle	
MRa329	A	1.04 g	KJo-0173 (extra dried)	1.56 g KOH (90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.43 g	shiny grey	crumpled sheets
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	42%	voluminous	
MB2990A	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.50 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	39%	brittle	
MB2990B	B	1.04 g	MB2982	1.56 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.42 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	40%	brittle	

Experiment	Oven	Qty	Lot	Base		grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Appearance	Structure
MRa331A	B	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)		mill (5 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.64 g	shiny grey	mostly porous
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	50%	voluminous	chunks
MRa331B	A	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)		mill (5 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g	shiny grey	mostly porous
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%	voluminous	chunks
MRa333A	A	1.28 g	MB2985 (extra dried)	1.15 g KOH (90%)	0.77 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.21 g	shiny grey	crumpled sheets
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	17%	voluminous	
MRa333B	B	1.04 g	MB2985 (extra dried)	0.94 g KOH (90%)	0.62 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.26 g	shiny grey	crumpled sheets
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	25%	voluminous	
MRa334A	A	1.04 g	MB2985 (extra dried)	1.25 g KOH (90%)	0.31 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.48 g	shiny grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	46%	voluminous	sheets
MRa334B	B	0.88 g	MB2985 (extra dried)	1.06 g KOH (90%)	0.26 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	33 g 10% HCl	100 °C N2 stream	0.39 g	shiny grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	45%	voluminous	sheets
MB2993A	A	1.28 g	MB2982	1.15 g KOH (90%)	0.77 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.22 g	shiny grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	18%	voluminous	sheets
MB2993B	B	1.04 g	MB2982	0.94 g KOH (90%)	0.62 g NaOH (98%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.19 g	shiny grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	18%	voluminous	sheets
MB2995A	A	1.28 g	MB2983 (extra dried)	1.92 g KOH (90%)		mill (4 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.54 g	shiny grey	mostly porous
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	42%	voluminous	chunks
MB2995B	B	1.04 g	MB2983 (extra dried)	1.56 g KOH (90%)		mill (4 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.46 g	shiny grey	sheet/chunk mix
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	44%	voluminous	
TB1144A	A	1.04 g	KJo-0174	1.56 g KOH (90%)		ball mill (15 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	38 g 10% HCl	100 °C N2 stream	0.61 g	black/grey	inhomogeneous
		ground biochar (brown powder) NOT compacted									(+ water)	(atm. Pressure)	59%	somewhat shiny	fibres/chunks
TB1144B	B	1.28 g	KJo-0174	1.92 g KOH (90%)		ball mill (15 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	54 g 10% HCl	100 °C N2 stream	0.36 g	black/grey shiny	chunks/fragments,
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	28%		large pores
MB2997A	A	1.28 g	MB2985 (extra dried)	1.92 g KOH (90%)		mill (5 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.55 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	somewhat shiny	
MB2997B	B	1.04 g	MB2985 (extra dried)	1.56 g KOH (90%)		mill (5 min)	N2	3 °C/min	800 °C	1 h	40 g 10% HCl	100 °C N2 stream	0.45 g	black/grey	porous chunks
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	43%	somewhat shiny	
MB2999A	A	1.28 g	KJo-0174	1.92 g KOH (90%)		ball mill (25 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.68 g	shiny black/grey	mostly
		ground biochar (brown powder) compacted to two pellets of equal size									(+ water)	(atm. Pressure)	53%	voluminous	sheet/chunk mix

Experiment	Oven	Qty	Lot	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output	Appearance	Structure
MB2999B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (25 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.40 g	shiny black/grey	inhomogeneous
		ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	38%	voluminous	fibres/chunks
TB1146	A	1.28 g	MB2982	1.92 g KOH (90%)	mill (5 min)	N2	3 °C/min	800 °C	1 h	51 g 10% HCl	100 °C N2 stream	0.52 g	black/grey	porous chunks
		ground biochar/KOH mix (brown powder, MB2990) dried at 100 °C, then compacted to two pellets of equal size								(+ water)	(atm. Pressure)	41%	somewhat shiny	
TB1147	B	1.08 g	MB2983 (extra dried)	1.62 g KOH (90%)	mill (4 min)	N2	3 °C/min	800 °C	1 h	37 g 10% HCl	100 °C N2 stream	0.36 g	black/grey	mostly porous
		ground biochar/KOH mix (brown powder, MB2995) dried at 100 °C, then compacted to two pellets of equal size								(+ water)	(atm. Pressure)	33%	somewhat shiny	chunks
MB3001A	A	1.28 g	KJo-0174	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.49 g	shiny black/grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	38%	very voluminous	sheets
MB3001B	B	1.04 g	KJo-0174	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.48 g	shiny black/grey	mostly crumpled
		ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	46%	very voluminous	sheets
MB3003A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.57 g	black/grey	mostly porous
		freshly ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	45%	somewhat shiny	chunks
MB3003B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.44 g	shiny black/grey	mostly
		freshly ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	42%	voluminous	sheet/chunk mix
MB3004A	A	1.28 g	MB2986 (2% water)	1.92 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream	0.59 g	shiny black/grey	mostly crumpled
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	46%	very voluminous	sheets
MB3004B	B	1.04 g	MB2986 (2% water)	1.56 g KOH (90%)	ball mill (35 Hz, 80 sec)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream	0.41 g	shiny black/grey	mostly
		ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	39%	voluminous	sheet/chunk mix
TB1151A	A	1.40 g	KJo-0174	2.10 g KOH (90%)	ball mill (35 Hz, 10 min)	N2	3 °C/min	800 °C	1 h	51 g 10% HCl	100 °C N2 stream		black/grey brittle	
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	0%		
TB1151B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (35 Hz, 10 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream		black/grey brittle	
		ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	0%		
MB3007A	A	1.20 g	KJo-0174	1.80 g KOH (90%)	ball mill (35 Hz, 30 min)	N2	3 °C/min	800 °C	1 h	48 g 10% HCl	100 °C N2 stream			
		ground biochar (brown powder) compacted to two pellets of equal size								(+ water)	(atm. Pressure)	0%		
MB3007B	B	1.08 g	KJo-0174	1.62 g KOH (90%)	ball mill (35 Hz, 30 min)	N2	3 °C/min	800 °C	1 h	39 g 10% HCl	100 °C N2 stream			
		ground biochar (brown powder) NOT compacted								(+ water)	(atm. Pressure)	0%		